

Document Classification Performance Benchmark

Head-to-Head Evaluation: `~typesafe/jev-latest` (Decisions API) vs `nvidia/nemotron-3.5-lightning` (Chat Completions)

Total Documents 24 Multi-Format Files (12 Initial + 12 Lengthy)	Overall Speedup 4.28x Faster JEV: 1.68s vs LLM: 7.17s	Total Time Saved 131.88s (-76.6%) 40.25s vs 172.13s	Accuracy Score 100.0% Parity Both 24/24 Correct
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1. Combined Benchmark Results Across Dataset Categories

Metric / Performance Parameter	Initial Files (12)	Lengthy Files (12)	Combined Files (24)
JEV (~typesafe/jev-latest Decisions API)			
Mean Latency / Document	1.735 s	1.619 s	1.677 s
Median Latency / Document	1.682 s	1.269 s	1.575 s
Min Latency	0.630 s	0.640 s	0.630 s
Max Latency	4.419 s	4.853 s	4.853 s
Latency Std Dev	1.000 s	1.146 s	1.053 s
Total Processing Time	20.822 s	19.428 s	40.250 s
Throughput (docs/sec)	0.58 docs/sec	0.62 docs/sec	0.60 docs/sec
Classification Accuracy	100.0% (12/12)	100.0% (12/12)	100.0% (24/24)
LLM (nvidia/nemotron-3.5-lightning Chat Completions)			
Mean Latency / Document	4.831 s	9.514 s	7.172 s
Median Latency / Document	2.955 s	4.838 s	3.531 s
Min Latency	1.692 s	1.705 s	1.692 s
Max Latency	20.841 s	47.026 s	47.026 s
Latency Std Dev	5.353 s	13.241 s	10.162 s
Total Processing Time	57.970 s	114.163 s	172.133 s
Throughput (docs/sec)	0.21 docs/sec	0.11 docs/sec	0.14 docs/sec
Classification Accuracy	100.0% (12/12)	100.0% (12/12)	100.0% (24/24)
Comparative Performance & Efficiency			
Speedup Factor (JEV vs LLM)	2.78x faster	5.88x faster	4.28x faster
Time Saved by JEV	37.15s (-64.1%)	94.73s (-83.0%)	131.88s (-76.6%)
Accuracy Delta (JEV - LLM)	+0.0%	+0.0%	+0.0%

2. Individual Table 1: Initial Files (12 Documents)

Metric	JEV (Decisions API)	LLM (Chat Completions)	Advantage / Ratio
Dataset Sample Size	12 documents	12 documents	Identical
Total Elapsed Time	20.822 s	57.970 s	-37.15s (-64.1%)
Mean Latency / Doc	1.735 s	4.831 s	2.78x faster
Median Latency / Doc	1.682 s	2.955 s	1.76x faster
Min Latency	0.630 s	1.692 s	+1.062s
Max Latency	4.419 s	20.841 s	+16.422s
Latency Std Dev	1.000 s	5.353 s	5.4x lower variance
Throughput	0.58 docs/sec	0.21 docs/sec	2.8x higher
Classification Accuracy	100.0% (12/12)	100.0% (12/12)	100% Parity

3. Individual Table 2: Lengthy Files (12 Documents)

Metric	JEV (Decisions API)	LLM (Chat Completions)	Advantage / Ratio
Dataset Sample Size	12 documents	12 documents	Identical
Total Elapsed Time	19.428 s	114.163 s	-94.73s (-83.0%)
Mean Latency / Doc	1.619 s	9.514 s	5.88x faster
Median Latency / Doc	1.269 s	4.838 s	3.81x faster
Min Latency	0.640 s	1.705 s	+1.065s
Max Latency	4.853 s	47.026 s	+42.173s
Latency Std Dev	1.146 s	13.241 s	11.6x lower variance
Throughput	0.62 docs/sec	0.11 docs/sec	5.9x higher
Classification Accuracy	100.0% (12/12)	100.0% (12/12)	100% Parity

4. Individual Table 3: Combined Dataset (24 Documents)

Metric	JEV (Decisions API)	LLM (Chat Completions)	Advantage / Ratio
Dataset Sample Size	24 documents	24 documents	Identical
Total Elapsed Time	40.250 s	172.133 s	-131.88s (-76.6%)
Mean Latency / Doc	1.677 s	7.172 s	4.28x faster
Median Latency / Doc	1.575 s	3.531 s	2.24x faster
Min Latency	0.630 s	1.692 s	+1.062s
Max Latency	4.853 s	47.026 s	+42.173s
Latency Std Dev	1.053 s	10.162 s	9.6x lower variance
Throughput	0.60 docs/sec	0.14 docs/sec	4.3x higher
Classification Accuracy	100.0% (24/24)	100.0% (24/24)	100% Parity

5. Individual Table 4: Per-File Head-to-Head Detailed Breakdown

Filename	Type	Ground Truth	JEV Pred	JEV Time	LLM Pred	LLM Time	Faster Engine
appendix_notes_c.docx	Initial	Finance	Finance	0.630s	Finance	6.278s	JEV (+897%)
archive_entry_04.docx	Initial	Education	Education	0.844s	Education	20.841s	JEV (+2369%)
briefing_packet_09.pdf	Initial	Law	Law	0.838s	Law	3.364s	JEV (+301%)
dossier_part_4.pdf	Initial	Technology	Technology	1.056s	Technology	3.698s	JEV (+250%)
field_observation_14.txt	Initial	Education	Education	2.103s	Education	4.303s	JEV (+105%)
internal_memo_402.pdf	Initial	Finance	Finance	1.471s	Finance	2.178s	JEV (+48%)
metrics_packet_88.txt	Initial	Finance	Finance	4.419s	Finance	2.546s	LLM (+74%)
project_nexus_v1.docx	Initial	Technology	Technology	1.681s	Technology	1.692s	JEV (+1%)
record_log_771.txt	Initial	Technology	Technology	2.102s	Technology	1.698s	LLM (+24%)
section_b_draft.txt	Initial	Law	Law	1.683s	Law	2.257s	JEV (+34%)
status_review_nov.pdf	Initial	Education	Education	1.890s	Education	1.876s	LLM (+1%)
summary_digest_05.docx	Initial	Law	Law	2.105s	Law	7.239s	JEV (+244%)
appellate_brief_in_re_tech.txt	Lengthy	Law	Law	0.844s	Law	47.026s	JEV (+5472%)
cloud_infrastructure_blueprint.docx	Lengthy	Technology	Technology	0.855s	Technology	3.155s	JEV (+269%)
consolidated_financial_statement.docx	Lengthy	Finance	Finance	0.857s	Finance	8.215s	JEV (+859%)
curriculum_standards_2026.txt	Lengthy	Education	Education	0.640s	Education	5.757s	JEV (+800%)
cybersecurity_threat_model.pdf	Lengthy	Technology	Technology	1.067s	Technology	2.179s	JEV (+104%)
distributed_systems_architecture.txt	Lengthy	Technology	Technology	1.065s	Technology	6.222s	JEV (+484%)
faculty_symposium_proceedings.docx	Lengthy	Education	Education	4.853s	Education	3.920s	LLM (+24%)
master_services_agreement.docx	Lengthy	Law	Law	1.679s	Law	23.676s	JEV (+1310%)
pedagogical_assessment_report.pdf	Lengthy	Education	Education	2.310s	Education	1.705s	LLM (+35%)
portfolio_risk_disclosure.pdf	Lengthy	Finance	Finance	1.472s	Finance	2.156s	JEV (+46%)
quarterly_treasury_audit.txt	Lengthy	Finance	Finance	1.889s	Finance	2.424s	JEV (+28%)
statutory_compliance_filing.pdf	Lengthy	Law	Law	1.897s	Law	7.728s	JEV (+307%)